

CS201 Midterm 1 Part B

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I. QUESTION (DOUBLYLINKEDLIST)

Write the following algorithm to sort the elements in the doubly linked list and return a new doubly linked list. The algorithm is as follows:

- Find the largest number N in the linked list.
- For each number i between 1 and N :
 - Count the number of times linked list has i .
 - Insert that many times i to the new linked list.

The elements before sorting:

5 2 3 4 2 3 4 3 2

The elements after sorting:

2 2 2 3 3 3 4 4 5

```
DoublyLinkedList sortElements()
```

You are not allowed to use any linked list methods. You are allowed to use attributes, constructors, getters and setters. Write the method in the `DoublyLinkedList` class.

II. QUESTION (LINKEDLIST)

Write the method

```
boolean evenOddSorted()
```

which returns true if the singly linked list odd indexed elements are sorted increasing order and even indexed elements are sorted in decreasing order. The first node has index 1. You are not allowed to use any singly linked list methods. You are allowed to use attributes, constructors, getters and setters. Write the method in the `LinkedList` class.

Sorted:

2 10 3 7 6 4 9 2 13 0 20

III. QUESTION (STACK)

Write a method that compresses the stack by removing the duplicate items. Assume that the elements in the stack are sorted. You are only allowed to use pop, push, isEmpty functions. Write the method for one of the array or linked list implementations. (Hint: Use external stack).

```
void compress()
```

Contents of the stack before:

1 1 2 2 2 5 5 5 6

Contents of the stack after:

1 2 5 6

IV. QUESTION (STACK)

Write the method

```
void push(int k, int data)
```

which pushes data as the k 'th element from the top. `push(1, data)` is equal to the original push, that is, the first element at the top has index 1. You are not allowed to use any stack methods and any external stacks. You are allowed to use attributes, constructors, getters and setters. Write the method for both array and linked list implementations.

Contents of the stack before:

1 2 0 4 6 9

Contents of the stack after `push(3, 10)`:

1 2 0 4 10 6 9

Contents of the stack after `push(5, 7)`:

1 2 0 7 4 10 6 9